

Feature bands study of MI signals based on FBCSP for CNN

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Abstract—Deep learning models have received a lot of attention from researchers in the field of brain-computer interface (BCI) due to their successful applications in the field of natural language processing as well as computer vision. The research shows that the convolutional neural network can be effectively applied to the classification and recognition of motion imagery (MI) signals, and has achieved good results. How to extract the spatial domain and time-frequency domain features of EEG signals and how to construct a sample matrix suitable for network training have been the key research problems in the field of EEG classification. In this paper, we start from the existing model and conduct experiments on the BCI public data set BCI Competition Set IV2a to investigate the feature bands of signals in different frequency band ranges, to construct the optimal feature matrix to improve the classification accuracy of the model. The experiments compare two traditional algorithms and an algorithm applying an artificial neural network.

Keywords—Brain-Computer Interface (BCI), Deep Learning (DL), Convolutional Neural Network (CNN), Motor Imagery Signal (MI), Feature Bands

I. INTRODUCTION

BCI is a class of communication devices that connect the brain to external devices independent of peripheral neural pathways [1], and their development has led to applications in several fields [2]. There are three types of EEG signals commonly used in BCI devices, which are steady-state visual evoked potentials, motor imagery signals, and P300 signals, among which motion imagery signals are a kind of EEG signals that generate without relying on external device stimulation.

The EEG signal object studied in this paper is the motor imagery signals. The classification and recognition process for EEG data can be into the following four parts, data acquisition, feature extraction, feature selection, and classification. Common EEG data preprocessing can be carried out from three aspects: time domain, frequency domain, and spatial domain. Among them, filtering methods in the time domain include AR filtering, filtering methods in the frequency domain include Fourier transforms, and filtering methods in the time-frequency domain include short-time Fourier and wavelet transform. Herbert Ramoser et al. proposed the Common Spatial Pattern (CSP) [3], and Kai Keng Ang et al. proposed the Filter Bank Common Spatial Pattern (FBCSP) [4], which are methods for spatial domain filtering.

Some traditional machine learning algorithms for classification and recognition include linear discriminant

analysis (LDA) and support vector machines (SVM). These models constructed by machine learning algorithms with common spatial pattern algorithms were applied to MI signals classification and achieved a high accuracy rate. In 2010, Alexandre Baracht et al. proposed a classification algorithm different from the traditional algorithm Riemann geometric classifier [5]. When Riemann geometric classifier was proposed, its classification performance exceeded that of machine learning algorithms. However, with the application of deep learning in EEG classification, it and traditional algorithms both showed a backward trend in classification performance.

In 2012, the convolutional neural network (CNN) won the ImageNet competition [2] with better learning algorithms, faster computational resources, and large annotated data sets [6]. Thus, a class of deep learning algorithms represented by convolutional neural networks has attracted the attention of a large group of researchers in the field of EEG signal recognition. Since 2015, Some researchers have applied CNN to EEG classification with good results. Yang, H.J. proposed a CNN frequency-based complementary feature map selection scheme [7], and Kwon, O.Y. proposed a subject-independent framework based on deep convolutional neural networks [8]. Such models with deep learning algorithms applied have two different interpretations as follows. 1 The signal preprocessing part at the front end of the network input is the feature extraction, and the feature selection part of the EEG data whole network is the classification recognition part. 2 The network includes feature extraction, selection, and classification recognition.

Whether the classification models on traditional algorithmic processes or deep learning algorithms, one of the main reasons for their limited accuracy is that the feature extraction algorithms fail to extract feature information with high discrimination in the feature extraction stage. Since the extracted features are not discriminative, this leads to a low classification accuracy of the model. There are two solutions to this problem: 1. Extract the feature information with discrimination by designing and optimizing the feature extraction algorithm based on the traditional machine learning model 2. Extract the feature information with discrimination by designing the network structure based on the depth learning model and configuring the hyperparameter of the network.

Many scholars have proposed specific feature extraction algorithms for the time domain, frequency domain, and spatial domain, such as Xuguang Ma et al. proposed a new processing algorithm that can reduce the effect of individual differences on classification and improve the generalization

of the model [9]. Based on the model of the article [10], this paper studies the influence of feature bands extracted under different frequency bands on subsequent classification.

II. DATA

The data for the nine subjects used in this paper were from BCI Competition Set IV2a [11], and the type of EEG signal generated was motor imagery signals. The cause of motor imagery signals can be traced back to a kind of electrophysiological phenomenon, that is, when subjects are engaged in motor imagination, the corresponding motor perception cortex of the brain will produce a potential rise, which is called event-related synchronization (ERS) phenomenon, and a potential decline, which is called event-related desynchronization (ERD) phenomenon [12]. The experimental paradigm is as follows. At first, the subjects sat quietly on the armchair in front of the computer screen. At the beginning of the experiment (0s), a fixed cross appeared on the screen, and a short warning sound was issued to prompt the subjects to start the trial, and two seconds later, an arrow representing one of the four categories appeared on the screen to prompt the subjects to engage in motor imagery until 6s, when the cross disappeared from the screen, during which the cross remained on the screen, and the subjects were asked to maintain motor imagery. Afterward, subjects went into a short rest for the subsequent trial. The EEG data were collected using 22 Ag/AgCl electrodes with an electrode spacing of 3.5 cm, and the sampling rate used in the experiment was 250 Hz. Evaluation of each model by the accuracy of this dataset after classification.

III. METHODS

The optimal frequency band feature extraction proposed in this paper is based on FBCSP, using Hilbert transform to extract the frequency domain features of EEG signals to achieve a generalized model and improve classification accuracy.

A. Extracting spatial features

The CSP algorithm uses the supervised method to create an optimal common spatial filter, maximizing one type of variance while minimizing the other, and diagonalizing the covariance matrix of two classes of tasks simultaneously to obtain the feature vector with the most degree of discrimination. The CSP algorithm flow is in Figure 1.

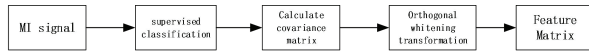


Figure 1. CSP algorithm flow diagram

The process of the CSP algorithm is as follows.

(1) Classify raw data according to supervised signals. The two categories of sample data E can be classified as E_1 and E_2 . E_1 is the first class of sample data, and E_2 is the second class of sample data. In the two classifications of motion imagery signals, one class of signals represents the left-hand motion imagery signals, and another class of signals represents the right-hand motion imagery signals.

(2) Calculate the covariance matrix of the raw data after classification. The covariance matrix calculation formula is defined as follows:

$$C_i = \frac{E_i \cdot E_i^T}{\text{trace}(E_i \cdot E_i^T)} (i=1,2) \quad (1)$$

$\text{trace}(E)$ denotes the trace of matrix E . Calculating the covariance matrix of the raw data after classification separately, C_1 is the expectation of the spatial covariance matrix of the first type of sample data, C_2 denotes the expectation of the spatial covariance matrix of the second type of sample, and C_c is the sum of the spatial covariance matrix of the two classes of data.

$$C_c = C_1 + C_2 \quad (2)$$

(3) The orthogonal whitening transformation is computed and diagonalized simultaneously, C_c is the positive definite matrix, and the singular value decomposition theorem was given by

$$C_c = U_c \Lambda_c U_c^T \quad (3)$$

U_c is the eigenvector matrix, Λ_c denotes the diagonal matrix of eigenvalues, and the eigenvalues are in descending order. The whitening transformation U_c gives:

$$P = \frac{1}{\sqrt{\Lambda_c}} \cdot U_c^T \quad (4)$$

Apply the matrix P to C_1 , C_2 :

$$S_1 = PC_1P^T, S_2 = PC_2P^T \quad (5)$$

S_1 , S_2 have common eigenvectors and two diagonal matrices Λ_1 , Λ_2 and eigenvector matrix B , satisfying the following conditions.

$$S_1 = B\Lambda_1B^T \quad (6)$$

$$S_2 = B\Lambda_2B^T \quad (7)$$

$$\Lambda_1 + \Lambda_2 = I \quad (8)$$

where I is the identity matrix. It shows the sum of the eigenvalues of S_1 and S_2 is 1.

(4) Calculate the projection matrix. For the eigenvector matrix B , when one category S_1 is the largest eigenvalue, the other category S_1 is the smallest eigenvalue, the matrix B can solve the binary classification problem, and the projection matrix given by

$$W = (B^T P)^T \quad (9)$$

(5) The raw EEG data $E_{M \times N}$ is projected through the projection matrix W to get the feature matrix.

$$Z_{M \times N} = W_{M \times M} E_{M \times N} \quad (10)$$

We can select the first m rows and the last m rows of $Z_{M \times N}$ ($2m < M$) as the characteristics of the original input data.

(6) Feature normalization

$$y_i = \log \left(\frac{\text{var}(Z_i)}{\sum_{n=1}^{2m} \text{var}(Z_n)} \right) \quad (11)$$

where y_i is the normalized feature matrix of the i th sample. The CSP feature matrix generated by the algorithm is not equivalent in terms of information. The feature information is mainly concentrated in the head and tail of the feature matrix, while the feature information in the middle is not obvious. Thus we can choose the data of m rows and the next m rows as the feature matrix for CSP feature extraction.

B. Extraction of frequency band features

a) We have conducted an in-depth study on the following four groups of band divisions: ① 0-40Hz, the division interval is 8Hz bandwidth ② 0-40Hz, the division interval is 4Hz bandwidth ③ 0-40Hz, the division interval is 2Hz bandwidth ④ 0-40Hz, the division interval is 1Hz bandwidth. In the following description, we will explain the second case. We take the period of motion imagination in the time dimension, i.e., the second to the sixth second after the start of the trial, with a total duration of four seconds, and we use a sampling frequency of 250 Hz. Then we obtain 1000 sample points of the MI signals in the time dimension. We segment the same signals for filtering. After that, we will get ten sets of filtered signals. We record it as [10,22,1000].

b) Since CSP is for binary classification, in this paper, there are four categories for each group of signals. The multi-classification strategies are one-to-one, one-to-the-rest, and many-to-many, where one-to-one and one-to-the-rest are cases of many-to-many. In this paper, we choose the one-to-the-rest multi-categorization. So, we will output four classifiers in the four classifications as follows. 1. The left hand is a category, and the right hand, the foot, and the tongue are a category. 2. The right hand is a category, and the left hand, the foot, and the tongue are a category. 3. The foot is a category, and the left hand, the right hand, and the tongue are a category. 4. The tongue is a category, and the left hand, the right hand, and the foot are a category. Ten groups of filtered signals, each group outputs four CSP classifiers are 10×4 CSP filter matrices, the number of channels of each matrix is the number of electrodes that is 22, According to the CSP algorithm in A, so. We obtain a filter matrix of dimension four, denoted as [10,4,22,22].

c) We filter the matrix signal [10, 22,1000] through the filter matrix [10, 4, 22, 22]. Note that this refers to the corresponding filtering of different classes of signals in each frequency band. After that, we will get a matrix signal of size $10 \times 4 \times 22 \times 1000$ with dimension four. We take the maximized and minimized pair of features to prepare for the subsequent feature selection. Take the first and the last of the 22 signals with 1000 samples, and do the modulo operation. Thus we get 80 features.

d) We use the feature selection method of mutual information for feature channel selection. In this paper, we conduct each set of experiments by choosing a base $k = 4$. Then the case of the maximum number of feature channels is

$k * m * c$, where m is equal to 2, representing the maximum and minimum feature vectors corresponding to each filter, and c is the number of categories, which is 4. Therefore, the maximum number of feature channels is 32.

C. Temporal Filtering

When we perform the third step of channel filtering, the feature channels are selected according to the mutual information feature selection, while the whole time filtering can into the following steps.

a) We perform the Hilbert transform on the signal in each band and then take it to its analytic signal. We note that the bandwidth of the raw signal is 40Hz. After the segmented filtering, the bandwidth of the ten signals is 4Hz, so the bandwidth of the raw signal's analytic signal is also 4Hz. We take the complex envelope of the ten analytic signals, and the signals change from Band-pass to Low-pass signals and the bandwidth changes from 4Hz to 2Hz.

b) In this way, we obtain ten low-pass signals with a cutoff frequency of 2 Hz, whose amplitudes we take directly, without other processing, as signals for subsequent processing.

c) According to the Nyquist sampling theorem, we use a frequency greater than $2 \times 2\text{Hz}$ for downsampling to obtain the raw signal information without distortion. We use a sampling frequency of 10Hz for downsampling. Then we get the sample matrix signal and record it as [32,40].

After the above three steps, we will have extracted sufficient time information from the time series of the raw signal and compressed the signal from the time dimension.

D. Design of convolutional neural network

In the design of convolutional neural networks, we refer to the C2CM model in this paper [13]. The model divides into three convolutional layers. The first two layers process the temporal features, and the third layer processes the channel features. The entire classification algorithm flow framework in Fig. 2. The visualization of the CNN model in Fig. 3.

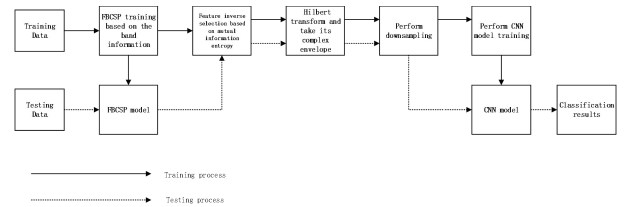


Figure 2. Processing flow of multi-category framework

E. Tenfold cross-validation

In the tenfold cross-validation, we selected the cross-entropy function as the loss function and the Adam function as the optimizer based on a priori knowledge [14].

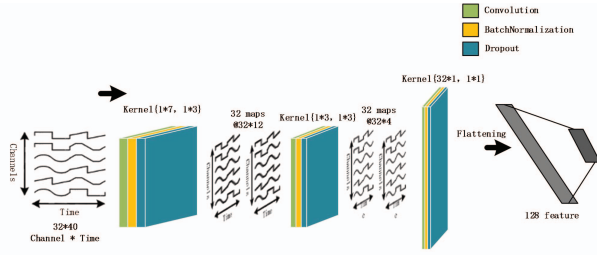


Figure 3. Convolutional neural network model visualization

IV. RESULTS

A. Baseline Methods

Three comparison algorithms in this paper. 1. Common spatial pattern plus linear discriminant analysis. 2. Common spatial pattern plus support vector machine. 3. Wavelet packet decomposition, common spatial pattern plus artificial neural network. The three algorithms carry out under the public dataset BCI competition set IV 2a, which includes nine subjects. The classification results are in Table I, in which the accuracy of each algorithm gives. We can see from Table I that the classification accuracy obtained by the artificial neural network approach is the highest, reaching 61.073%. The worst result was obtained by the common spatial model plus linear discriminant analysis, with a classification accuracy of 41.609%. Our C2CM model resulted in a classification accuracy of 58.966%. Another traditional classification method, the common spatial pattern and support vector machine, achieved a classification accuracy of 43.793%.

TABLE I. COMPARISON OF THE CLASSIFICATION ACCURACY OF FOUR TYPES OF METHODS

Methods	CSP+LDA	CSP+SVM	WPD+CSP+ANN	C2CM
subject1	63.10%	58.97%	72.07%	82.41%
subject2	25.86%	28.62%	61.03%	53.79%
subject3	59.31%	52.76%	84.83%	82.76%
subject4	26.55%	37.59%	43.10%	37.59%
subject5	25.86%	27.59%	52.76%	40.00%
subject6	29.66%	36.90%	40.69%	46.90%
subject7	63.79%	63.10%	76.21%	79.31%
subject8	40.69%	43.45%	72.07%	48.97%
subject9	39.66%	45.17%	46.90%	58.97%
average	41.61%	43.79%	61.07%	58.97%

B. Comparison under different frequency bands

In this paper, based on the model of C2CM, we studied the influence of the extracted features on the classification results under different frequency bands. We used four different sets of frequency band intervals for the experiments. The experimental results are in Table 2. In Table II, we can see that the average classification accuracy is 58.966% in the third group as the control group, i.e., the control group with the frequency band divided into ten groups. In the first group, the experimental group that divided the frequency band into 40 groups, the classification accuracy reached the highest, which was 62.797%. In the second group, i.e., the experimental group whose frequency bands divide into 20

fourth group, i.e., the experimental group in which the frequency bands divide into five groups, the classification accuracy was only 41.111%.

C. Comparison of improved model and artificial neural network model

The comparison between the improved model and the artificial neural network model is in Table III. The model with the optimal frequency band performs slightly lower than the ANN model on individual subjects. Overall the average classification accuracy of the improved model is higher than that of the ANN model.

TABLE II. COMPARISON OF THE CLASSIFICATION ACCURACIES OBTAINED AT DIFFERENT BAND WIDTHS

Bandwidth	0-40-1	0-40-2	0-40-4	0-40-8
subject1	86.21%	81.72%	82.41%	27.93%
subject2	52.76%	56.90%	53.79%	44.48%
subject3	85.52%	83.10%	82.76%	69.66%
subject4	39.31%	37.59%	37.59%	34.48%
subject5	52.76%	35.52%	40.00%	28.97%
subject6	47.59%	38.28%	46.90%	35.86%
subject7	72.41%	66.21%	79.31%	51.72%
subject8	53.45%	34.48%	48.97%	32.76%
subject9	75.17%	46.55%	58.97%	44.14%
average	62.80%	53.37%	58.97%	41.11%

TABLE III. COMPARISON OF THE CLASSIFICATION ACCURACY WITH FOUR TYPES OF METHODS AT THE OBTAINED OPTIMAL FREQUENCY BAND

Methods	0-40-1	CSP+LD A	CSP+SV M	WPD+C SP+ANN	C2CM
subject1	86.21%	63.10%	58.97%	72.07%	82.41%
subject2	52.76%	25.86%	28.62%	61.03%	53.79%
subject3	85.52%	59.31%	52.76%	84.83%	82.76%
subject4	39.31%	26.55%	37.59%	43.10%	37.59%
subject5	52.76%	25.86%	27.59%	52.76%	40.00%
subject6	47.59%	29.66%	36.90%	40.69%	46.90%
subject7	72.41%	63.79%	63.10%	76.21%	79.31%
subject8	53.45%	40.69%	43.45%	72.07%	48.97%
subject9	75.17%	39.66%	45.17%	46.90%	58.97%
average	62.80%	41.61%	43.79%	61.07%	58.97%

V. DISCUSSIONS

We can see from Tables I and II the fact that subjects 1, 3, and 7 performed relatively stable and good classification results when assessed by various methods, while subjects 2, 4, 5, and 6 all performed poorly. The above analysis confirms the individual variability and non-stationarity of MI signals generated by electrophysiological phenomena such as ERD and ERS.

After we further analyze the results in Table II, we can find that when the bandwidth is set to 8 Hz and divided into five groups, the classification accuracy is much lower than that of the control group, which is set to 4 Hz and divided into ten groups, by 17.8544%. It indicates that making a larger width setting for the frequency band will make the feature information in the band redundant, which is not conducive to feeding into the CNN network for classification and recognition. When we set the frequency bandwidth to 2Hz and divided it into 20 groups, the classification accuracy

decreased slightly compared with the control group, which can be explained by the loss of part of feature information, while the information components with high discrimination still have redundancy and fail to be extracted. When we set the frequency bandwidth to 1Hz and divide it into 40 groups, the classification accuracy is higher than that of the control group and higher than that of the group using the artificial neural network. It shows that although the frequency bandwidth becomes smaller and some feature information is lost, the feature information with high discrimination is highlighted, which makes the extracted feature information easy to send to CNN for training. At the same time, The classification results also well confirm this point. In addition, from Figure 4, we can also see that the CNN model trained by using the optimal frequency band is also better than the original model in fitting the loss curve, which shows that the generalization of the improved model has been improved.

In the following figure, the right side is the improved loss diagram of the model.

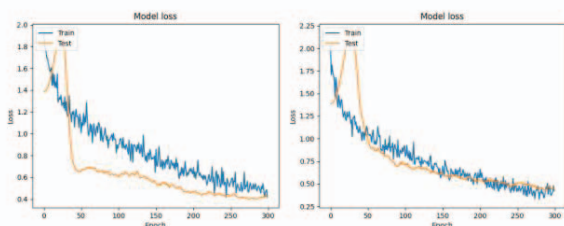


Figure 4. Comparison of the loss curve with the original model at the optimal frequency band

VI. SUMMARY AND PROSPECTS

In this paper, firstly, the C2CM model, two traditional classification algorithms, and a deep learning method are used to conduct a comparison experiment with the same data set. The experimental results show that the classification results of C2CM are similar but slightly lower, the best deep learning algorithm among the three compared algorithms. As a result, the range of feature bands used for CNN based on FBCSP is investigated in this paper. It shows that, with the total bandwidth unchanged, narrowing the divided band bandwidth, eliminating redundant information, and retaining the most differentiated band information are more conducive to feeding into CNN for training models and subsequent classification recognition. Using the improved feature band, the classification result is better than the original algorithm and the algorithm using an artificial neural network model.

This paper only studies the impact of frequency bands within the information range of feature frequency bands divided according to different frequency bands on the results and does not study other more complex situations. Therefore, our future research work can focus on ① the research under the condition of an uneven division of frequency band bandwidth. ② The study of overlapping bandwidth division.

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